Types of Morphisms in Bicategories

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In this chapter, we study special kinds of morphisms in bicategories:

1. *Monomorphisms and Epimorphisms in Bicategories (Sections 13.1 and 13.2).* There is a large number of different notions capturing the idea of a "monomorphism" or of an "epimorphism" in a bicategory.

   Arguably, the notion that best captures these concepts is that of a *pseudomonic morphism* (Definition 13.1.10.1.1) and of a *pseudoepic morphism* (Definition 13.2.10.1.1), although the other notions introduced in Sections 13.1 and 13.2 are also interesting on their own.

Contents

13.1 Monomorphisms in Bicategories **2**
13.1.1 Representably Faithful Morphisms 2
13.1.2 Representably Full Morphisms 3
13.1.3 Representably Fully Faithful Morphisms 4
13.1.4 Morphisms Representably Faithful on Cores 5
13.1.5 Morphisms Representably Full on Cores 6
13.1.6 Morphisms Representably Fully Faithful on Cores 7
13.1.7 Representably Essentially Injective Morphisms 8
13.1.8 Representably Conservative Morphisms 8
13.1.9 Strict Monomorphisms 9
13.1.10 Pseudomonic Morphisms 10

13.2 Epimorphisms in Bicategories **11**
13.2.1 Corepresentably Faithful Morphisms 11
13.2.2 Corepresentably Full Morphisms 12

13.2.3 Corepresentably Fully Faithful Morphisms 13
13.2.4 Morphisms Corepresentably Faithful on Cores 15
13.2.5 Morphisms Corepresentably Full on Cores 15
13.2.6 Morphisms Corepresentably Fully Faithful on Cores 16
13.2.7 Corepresentably Essentially Injective Morphisms 18
13.2.8 Corepresentably Conservative Morphisms 18
13.2.9 Strict Epimorphisms 19
13.2.10 Pseudoepic Morphisms 19

A Other Chapters 21

13.1 Monomorphisms in Bicategories

13.1.1 Representably Faithful Morphisms

Let $\mathcal{C}$ be a bicategory.

Definition 13.1.1.1.1. A 1-morphism $f\colon A \to B$ of $\mathcal{C}$ is **representably faithful**[1] if, for each $X \in \mathrm{Obj}(\mathcal{C})$, the functor

$$f_*\colon \mathsf{Hom}_{\mathcal{C}}(X, A) \to \mathsf{Hom}_{\mathcal{C}}(X, B)$$

given by postcomposition by f is faithful.

Remark 13.1.1.1.2. In detail, f is representably faithful if, for all diagrams in $\mathcal{C}$ of the form

$$\begin{tikzcd} X \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\alpha\Downarrow\Downarrow\beta"] & A \arrow[r, "f"] & B, \end{tikzcd}$$

if we have

$$\mathrm{id}_f \star \alpha = \mathrm{id}_f \star \beta,$$

then $\alpha = \beta$.

Example 13.1.1.1.3. Here are some examples of representably faithful morphisms.

1. *Representably Faithful Morphisms in* Cats_2. The representably faithful morphisms in Cats_2 are precisely the faithful functors; see Categories, Item 2 of Definition 11.6.1.1.2.

[1] *Further Terminology:* Also called simply a **faithful morphism**, based on Item 1 of

2. *Representably Faithful Morphisms in* **Rel**. Every morphism of **Rel** is representably faithful; see Relations, Item 1 of Definition 8.4.8.1.1.

13.1.2 Representably Full Morphisms

Let $\mathcal{C}$ be a bicategory.

Definition 13.1.2.1.1. A 1-morphism $f\colon A \to B$ of $\mathcal{C}$ is **representably full**[2] if, for each $X \in \mathrm{Obj}(\mathcal{C})$, the functor

$$f_*\colon \mathsf{Hom}_{\mathcal{C}}(X, A) \to \mathsf{Hom}_{\mathcal{C}}(X, B)$$

given by postcomposition by f is full.

Remark 13.1.2.1.2. In detail, f is representably full if, for each $X \in \mathrm{Obj}(\mathcal{C})$ and each 2-morphism

$$\beta\colon f \circ \phi \Longrightarrow f \circ \psi, \qquad \begin{tikzcd} X \arrow[r, bend left=60, "f\circ\phi"{name=U}] \arrow[r, bend right=60, "f\circ\psi"'{name=D}] & B \arrow[Rightarrow, from=U, to=D, "\beta"] \end{tikzcd}$$

of $\mathcal{C}$, there exists a 2-morphism

$$\alpha\colon \phi \Longrightarrow \psi, \qquad \begin{tikzcd} X \arrow[r, bend left=60, "\phi"{name=U}] \arrow[r, bend right=60, "\psi"'{name=D}] & A \arrow[Rightarrow, from=U, to=D, "\alpha"] \end{tikzcd}$$

of $\mathcal{C}$ such that we have an equality

$$\begin{tikzcd} X \arrow[r, bend left=60, "\phi"{name=U}] \arrow[r, bend right=60, "\psi"'{name=D}] & A \arrow[Rightarrow, from=U, to=D, "\alpha"] \arrow[r, "f"] & B \end{tikzcd} \quad = \quad \begin{tikzcd} X \arrow[r, bend left=60, "f\circ\phi"{name=U}] \arrow[r, bend right=60, "f\circ\psi"'{name=D}] & B \arrow[Rightarrow, from=U, to=D, "\beta"] \end{tikzcd}$$

of pasting diagrams in $\mathcal{C}$, i.e. such that we have

$$\beta = \mathrm{id}_f \star \alpha.$$

Example 13.1.2.1.3. Here are some examples of representably full morphisms.

Definition 13.1.1.1.3.

[2] *Further Terminology:* Also called simply a **full morphism**, based on Item 1 of

1. *Representably Full Morphisms in* Cats_2. The representably full morphisms in Cats_2 are precisely the full functors; see Categories, **??** of Definition 11.6.2.1.2.

2. *Representably Full Morphisms in* **Rel**. The representably full morphisms in **Rel** are characterised in Relations, Item 2 of Definition 8.4.8.1.1.

13.1.3 Representably Fully Faithful Morphisms

Let $\mathcal{C}$ be a bicategory.

Definition 13.1.3.1.1. A 1-morphism $f\colon A \to B$ of $\mathcal{C}$ is **representably fully faithful**[3] if the following equivalent conditions are satisfied:

1. The 1-morphism f is representably faithful (Definition 13.1.1.1.1) and representably full (Definition 13.1.2.1.1).

2. For each $X \in \mathrm{Obj}(\mathcal{C})$, the functor

$$f_*\colon \mathsf{Hom}_{\mathcal{C}}(X, A) \to \mathsf{Hom}_{\mathcal{C}}(X, B)$$

given by postcomposition by f is fully faithful.

Remark 13.1.3.1.2. In detail, f is representably fully faithful if the conditions in Definition 13.1.1.1.2 and Definition 13.1.2.1.2 hold:

1. For all diagrams in $\mathcal{C}$ of the form

$$\begin{tikzcd} X \arrow[r, bend left, "\phi"{name=U}] \arrow[r, bend right, "\psi"'{name=D}] & A \arrow[r, "f"] & B, \arrow[Rightarrow, from=U, to=D, "\alpha"'] \arrow[Rightarrow, from=U, to=D, "\beta"] \end{tikzcd}$$

if we have

$$\mathrm{id}_f \star \alpha = \mathrm{id}_f \star \beta,$$

then $\alpha = \beta$.

Definition 13.1.2.1.3.

[3] *Further Terminology:* Also called simply a **fully faithful morphism**, based on Item 1 of Definition 13.1.3.1.3.

2. For each $X \in \mathrm{Obj}(\mathcal{C})$ and each 2-morphism

$$\beta\colon f \circ \phi \Longrightarrow f \circ \psi, \qquad \begin{tikzcd} X \arrow[r, bend left=60, "f\circ\phi"{name=U}] \arrow[r, bend right=60, "f\circ\psi"'{name=D}] & B \arrow[Rightarrow, from=U, to=D, "\beta"] \end{tikzcd}$$

of $\mathcal{C}$, there exists a 2-morphism

$$\alpha\colon \phi \Longrightarrow \psi, \qquad \begin{tikzcd} X \arrow[r, bend left=60, "\phi"{name=U}] \arrow[r, bend right=60, "\psi"'{name=D}] & A \arrow[Rightarrow, from=U, to=D, "\alpha"] \end{tikzcd}$$

of $\mathcal{C}$ such that we have an equality

$$\begin{tikzcd} X \arrow[r, bend left=60, "\phi"{name=U}] \arrow[r, bend right=60, "\psi"'{name=D}] & A \arrow[r, "f"] \arrow[Rightarrow, from=U, to=D, "\alpha"] & B \end{tikzcd} \quad = \quad \begin{tikzcd} X \arrow[r, bend left=60, "f\circ\phi"{name=U}] \arrow[r, bend right=60, "f\circ\psi"'{name=D}] & B \arrow[Rightarrow, from=U, to=D, "\beta"] \end{tikzcd}$$

of pasting diagrams in $\mathcal{C}$, i.e. such that we have

$$\beta = \mathrm{id}_f \star \alpha.$$

Example 13.1.3.1.3. Here are some examples of representably fully faithful morphisms.

1. *Representably Fully Faithful Morphisms in* Cats_2. The representably fully faithful morphisms in Cats_2 are precisely the fully faithful functors; see Categories, Item 6 of Definition 11.6.3.1.2.

2. *Representably Fully Faithful Morphisms in* **Rel**. The representably fully faithful morphisms of **Rel** coincide (Relations, Item 3 of Definition 8.4.8.1.1) with the representably full morphisms in **Rel**, which are characterised in Relations, Item 2 of Definition 8.4.8.1.1.

13.1.4 Morphisms Representably Faithful on Cores

Let $\mathcal{C}$ be a bicategory.

Definition 13.1.4.1.1. A 1-morphism $f\colon A \to B$ of $\mathcal{C}$ is **representably faithful on cores** if, for each $X \in \mathrm{Obj}(\mathcal{C})$, the functor

$$f_*\colon \mathsf{Core}(\mathsf{Hom}_{\mathcal{C}}(X, A)) \to \mathsf{Core}(\mathsf{Hom}_{\mathcal{C}}(X, B))$$

given by postcomposition by f is faithful.

Remark 13.1.4.1.2. In detail, f is representably faithful on cores if, for all diagrams in $\mathcal{C}$ of the form

$$\begin{tikzcd} X \arrow[r, bend left=60, "\phi"{name=U}] \arrow[r, bend right=60, "\psi"'{name=D}] & A \arrow[r, "f"] & B \arrow[from=U, to=D, Rightarrow, "\alpha"] \arrow[from=U, to=D, Rightarrow, "\beta"'] \end{tikzcd},$$

if α and β are 2-isomorphisms and we have

$$\mathrm{id}_f \star \alpha = \mathrm{id}_f \star \beta,$$

then $\alpha = \beta$.

13.1.5 Morphisms Representably Full on Cores

Let $\mathcal{C}$ be a bicategory.

Definition 13.1.5.1.1. A 1-morphism $f\colon A \to B$ of $\mathcal{C}$ is **representably full on cores** if, for each $X \in \mathrm{Obj}(\mathcal{C})$, the functor

$$f_*\colon \mathsf{Core}(\mathsf{Hom}_{\mathcal{C}}(X, A)) \to \mathsf{Core}(\mathsf{Hom}_{\mathcal{C}}(X, B))$$

given by postcomposition by f is full.

Remark 13.1.5.1.2. In detail, f is representably full on cores if, for each $X \in \mathrm{Obj}(\mathcal{C})$ and each 2-isomorphism

$$\beta\colon f \circ \phi \overset{\sim}{\Longrightarrow} f \circ \psi, \qquad \begin{tikzcd} X \arrow[r, bend left=60, "f\circ\phi"{name=U}] \arrow[r, bend right=60, "f\circ\psi"'{name=D}] & B \arrow[from=U, to=D, Rightarrow, "\beta"] \end{tikzcd}$$

of $\mathcal{C}$, there exists a 2-isomorphism

$$\alpha\colon \phi \overset{\sim}{\Longrightarrow} \psi, \qquad \begin{tikzcd} X \arrow[r, bend left=60, "\phi"{name=U}] \arrow[r, bend right=60, "\psi"'{name=D}] & A \arrow[from=U, to=D, Rightarrow, "\alpha"] \end{tikzcd}$$

of $\mathcal{C}$ such that we have an equality

$$\begin{tikzcd} X \arrow[r, bend left=60, "\phi"{name=U}] \arrow[r, bend right=60, "\psi"'{name=D}] & A \arrow[r, "f"] & B \arrow[from=U, to=D, Rightarrow, "\alpha"] \end{tikzcd} \quad = \quad \begin{tikzcd} X \arrow[r, bend left=60, "f\circ\phi"{name=U}] \arrow[r, bend right=60, "f\circ\psi"'{name=D}] & B \arrow[from=U, to=D, Rightarrow, "\beta"] \end{tikzcd}$$

of pasting diagrams in $\mathcal{C}$, i.e. such that we have

$$\beta = \mathrm{id}_f \star \alpha.$$

13.1.6 Morphisms Representably Fully Faithful on Cores

Let $\mathcal{C}$ be a bicategory.

Definition 13.1.6.1.1. A 1-morphism $f\colon A \to B$ of $\mathcal{C}$ is **representably fully faithful on cores** if the following equivalent conditions are satisfied:

1. The 1-morphism f is representably faithful on cores (Definition 13.1.5.1.1) and representably full on cores (Definition 13.1.4.1.1).

2. For each $X \in \mathrm{Obj}(\mathcal{C})$, the functor

$$f_*\colon \mathsf{Core}(\mathsf{Hom}_{\mathcal{C}}(X, A)) \to \mathsf{Core}(\mathsf{Hom}_{\mathcal{C}}(X, B))$$

given by postcomposition by f is fully faithful.

Remark 13.1.6.1.2. In detail, f is representably fully faithful on cores if the conditions in Definition 13.1.4.1.2 and Definition 13.1.5.1.2 hold:

1. For all diagrams in $\mathcal{C}$ of the form

$$\begin{tikzcd} X \arrow[r, bend left=60, "\phi", "\psi"'] \arrow[r, phantom, "\alpha\Downarrow\Downarrow\beta"] & A \arrow[r, "f"] & B, \end{tikzcd}$$

if α and β are 2-isomorphisms and we have

$$\mathrm{id}_f \star \alpha = \mathrm{id}_f \star \beta,$$

then $\alpha = \beta$.

2. For each $X \in \mathrm{Obj}(\mathcal{C})$ and each 2-isomorphism

$$\beta\colon f \circ \phi \overset{\sim}{\Longrightarrow} f \circ \psi, \qquad \begin{tikzcd} X \arrow[r, bend left=60, "f\circ\phi", "f\circ\psi"'] \arrow[r, phantom, "\beta\Downarrow"] & B \end{tikzcd}$$

of $\mathcal{C}$, there exists a 2-isomorphism

$$\alpha\colon \phi \overset{\sim}{\Longrightarrow} \psi, \qquad \begin{tikzcd} X \arrow[r, bend left=60, "\phi", "\psi"'] \arrow[r, phantom, "\alpha\Downarrow"] & A \end{tikzcd}$$

of $\mathcal{C}$ such that we have an equality

$$\begin{tikzcd} X \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\alpha\Downarrow"] & A \arrow[r, "f"] & B \end{tikzcd} \quad = \quad \begin{tikzcd} X \arrow[r, bend left, "f\circ\phi"] \arrow[r, bend right, "f\circ\psi"'] \arrow[r, phantom, "\beta\Downarrow"] & B \end{tikzcd}$$

of pasting diagrams in $\mathcal{C}$, i.e. such that we have

$$\beta = \mathrm{id}_f \star \alpha.$$

13.1.7 Representably Essentially Injective Morphisms

Let $\mathcal{C}$ be a bicategory.

Definition 13.1.7.1.1. A 1-morphism $f\colon A \to B$ of $\mathcal{C}$ is **representably essentially injective** if, for each $X \in \mathrm{Obj}(\mathcal{C})$, the functor

$$f_*\colon \mathsf{Hom}_{\mathcal{C}}(X, A) \to \mathsf{Hom}_{\mathcal{C}}(X, B)$$

given by postcomposition by f is essentially injective.

Remark 13.1.7.1.2. In detail, f is representably essentially injective if, for each pair of morphisms $\phi, \psi\colon X \rightrightarrows A$ of $\mathcal{C}$, the following condition is satisfied:

($\star$) If $f \circ \phi \cong f \circ \psi$, then $\phi \cong \psi$.

13.1.8 Representably Conservative Morphisms

Let $\mathcal{C}$ be a bicategory.

Definition 13.1.8.1.1. A 1-morphism $f\colon A \to B$ of $\mathcal{C}$ is **representably conservative** if, for each $X \in \mathrm{Obj}(\mathcal{C})$, the functor

$$f_*\colon \mathsf{Hom}_{\mathcal{C}}(X, A) \to \mathsf{Hom}_{\mathcal{C}}(X, B)$$

given by postcomposition by f is conservative.

Remark 13.1.8.1.2. In detail, f is representably conservative if, for each pair of morphisms $\phi, \psi\colon X \rightrightarrows A$ and each 2-morphism

$$\alpha\colon \phi \Longrightarrow \psi, \qquad \begin{tikzcd} X \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\alpha\Downarrow"] & A \end{tikzcd}$$

of $\mathcal{C}$, if the 2-morphism

$$\mathrm{id}_f \star \alpha \colon f \circ \phi \Longrightarrow f \circ \psi, \qquad \begin{tikzcd} X \arrow[r, bend left=60, "f\circ\phi"{name=U}] \arrow[r, bend right=60, "f\circ\psi"'{name=D}] & B \arrow[from=U, to=D, Rightarrow, "\mathrm{id}_f \star \alpha"] \end{tikzcd}$$

is a 2-isomorphism, then so is α.

13.1.9 Strict Monomorphisms

Let $\mathcal{C}$ be a bicategory.

Definition 13.1.9.1.1. A 1-morphism $f \colon A \to B$ of $\mathcal{C}$ is a **strict monomorphism** if, for each $X \in \mathrm{Obj}(\mathcal{C})$, the functor

$$f_* \colon \mathsf{Hom}_{\mathcal{C}}(X, A) \to \mathsf{Hom}_{\mathcal{C}}(X, B)$$

given by postcomposition by f is injective on objects, i.e. its action on objects

$$f_* \colon \mathrm{Obj}(\mathsf{Hom}_{\mathcal{C}}(X, A)) \to \mathrm{Obj}(\mathsf{Hom}_{\mathcal{C}}(X, B))$$

is injective.

Remark 13.1.9.1.2. In detail, f is a strict monomorphism in $\mathcal{C}$ if, for each diagram in $\mathcal{C}$ of the form

$$\begin{tikzcd} X \arrow[r, shift left, "\phi"] \arrow[r, shift right, "\psi"'] & A \arrow[r, "f"] & B, \end{tikzcd}$$

if $f \circ \phi = f \circ \psi$, then $\phi = \psi$.

Example 13.1.9.1.3. Here are some examples of strict monomorphisms.

1. *Strict Monomorphisms in* Cats_2. The strict monomorphisms in Cats_2 are precisely the functors which are injective on objects and injective on morphisms; see Categories, Item 1 of Definition 11.7.2.1.2.

2. *Strict Monomorphisms in* **Rel**. The strict monomorphisms in **Rel** are characterised in Relations, Definition 8.4.7.1.1.

13.1.10 Pseudomonic Morphisms

Let $\mathcal{C}$ be a bicategory.

Definition 13.1.10.1.1. A 1-morphism $f\colon A \to B$ of $\mathcal{C}$ is **pseudomonic** if, for each $X \in \mathrm{Obj}(\mathcal{C})$, the functor

$$f_*\colon \mathsf{Hom}_{\mathcal{C}}(X, A) \to \mathsf{Hom}_{\mathcal{C}}(X, B)$$

given by postcomposition by f is pseudomonic.

Remark 13.1.10.1.2. In detail, a 1-morphism $f\colon A \to B$ of $\mathcal{C}$ is pseudomonic if it satisfies the following conditions:

1. For all diagrams in $\mathcal{C}$ of the form

$$\begin{tikzcd} X \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\alpha\Downarrow\Downarrow\beta"] & A \arrow[r, "f"] & B, \end{tikzcd}$$

   if we have

$$\mathrm{id}_f \star \alpha = \mathrm{id}_f \star \beta,$$

   then $\alpha = \beta$.

2. For each $X \in \mathrm{Obj}(\mathcal{C})$ and each 2-isomorphism

$$\beta\colon f \circ \phi \overset{\sim}{\Longrightarrow} f \circ \psi, \qquad \begin{tikzcd} X \arrow[r, bend left, "f\circ\phi"] \arrow[r, bend right, "f\circ\psi"'] \arrow[r, phantom, "\beta\Downarrow"] & B \end{tikzcd}$$

   of $\mathcal{C}$, there exists a 2-isomorphism

$$\alpha\colon \phi \overset{\sim}{\Longrightarrow} \psi, \qquad \begin{tikzcd} X \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\alpha\Downarrow"] & A \end{tikzcd}$$

   of $\mathcal{C}$ such that we have an equality

$$\begin{tikzcd} X \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\alpha\Downarrow"] & A \arrow[r, "f"] & B \end{tikzcd} \quad = \quad \begin{tikzcd} X \arrow[r, bend left, "f\circ\phi"] \arrow[r, bend right, "f\circ\psi"'] \arrow[r, phantom, "\beta\Downarrow"] & B \end{tikzcd}$$

   of pasting diagrams in $\mathcal{C}$, i.e. such that we have

$$\beta = \mathrm{id}_f \star \alpha.$$

Proposition 13.1.10.1.3. Let $f\colon A \to B$ be a 1-morphism of $\mathcal{C}$.

1. *Characterisations.* The following conditions are equivalent:

   (a) The morphism f is pseudomonic.

   (b) The morphism f is representably full on cores and representably faithful.

   (c) We have an isocomma square of the form

$$A \overset{\text{eq.}}{\cong} A \overset{\leftrightarrow}{\times}_B A, \qquad \begin{tikzcd} A \arrow[r, "\mathrm{id}_A"] \arrow[d, "\mathrm{id}_A"'] & A \arrow[d, "F"] \\ A \arrow[r, "F"'] \arrow[ur, Rightarrow, dashed] & B \end{tikzcd}$$

   in $\mathcal{C}$ up to equivalence.

2. *Interaction With Cotensors.* If $\mathcal{C}$ has cotensors with $\mathbb{1}$, then the following conditions are equivalent:

   (a) The morphism f is pseudomonic.

   (b) We have an isocomma square of the form

$$A \overset{\text{eq.}}{\cong} A \overset{\leftrightarrow}{\times}_{\mathbb{1} \pitchfork F} B, \qquad \begin{tikzcd} A \arrow[r, hook] \arrow[d, "F"'] & \mathbb{1} \pitchfork A \arrow[d, "\mathbb{1} \pitchfork F"] \\ B \arrow[r, hook] \arrow[ur, Rightarrow, dashed] & \mathbb{1} \pitchfork B \end{tikzcd}$$

   in $\mathcal{C}$ up to equivalence.

Proof. *Item 1*, *Characterisations*: Omitted.
Item 2, *Interaction With Cotensors*: Omitted. □

13.2 Epimorphisms in Bicategories

13.2.1 Corepresentably Faithful Morphisms

Let $\mathcal{C}$ be a bicategory.

Definition 13.2.1.1.1. A 1-morphism $f\colon A \to B$ of $\mathcal{C}$ is **corepresentably faithful** if, for each $X \in \mathrm{Obj}(\mathcal{C})$, the functor

$$f^*\colon \mathsf{Hom}_{\mathcal{C}}(B, X) \to \mathsf{Hom}_{\mathcal{C}}(A, X)$$

given by precomposition by f is faithful.

Remark 13.2.1.1.2. In detail, f is corepresentably faithful if, for all diagrams in $\mathcal{C}$ of the form

$$\begin{tikzcd} A \arrow[r, "f"] & B \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\alpha\Downarrow\Downarrow\beta"] & X, \end{tikzcd}$$

if we have

$$\alpha \star \mathrm{id}_f = \beta \star \mathrm{id}_f,$$

then $\alpha = \beta$.

Example 13.2.1.1.3. Here are some examples of corepresentably faithful morphisms.

1. *Corepresentably Faithful Morphisms in* Cats_2. The corepresentably faithful morphisms in Cats_2 are characterised in Categories, Item 5 of Definition 11.6.1.1.2.

2. *Corepresentably Faithful Morphisms in* **Rel**. Every morphism of **Rel** is corepresentably faithful; see Relations, Item 1 of Definition 8.4.10.1.1.

13.2.2 Corepresentably Full Morphisms

Let $\mathcal{C}$ be a bicategory.

Definition 13.2.2.1.1. A 1-morphism $f\colon A \to B$ of $\mathcal{C}$ is **corepresentably full** if, for each $X \in \mathrm{Obj}(\mathcal{C})$, the functor

$$f^*\colon \mathsf{Hom}_{\mathcal{C}}(B, X) \to \mathsf{Hom}_{\mathcal{C}}(A, X)$$

given by precomposition by f is full.

Remark 13.2.2.1.2. In detail, f is corepresentably full if, for each $X \in$

$\mathrm{Obj}(\mathcal{C})$ and each 2-morphism

$$\beta\colon \phi \circ f \Longrightarrow \psi \circ f, \qquad \begin{tikzcd} A \arrow[r, bend left=60, "\phi\circ f"{name=U}] \arrow[r, bend right=60, "\psi\circ f"'{name=D}] & X \arrow[Rightarrow, from=U, to=D, "\beta"] \end{tikzcd}$$

of $\mathcal{C}$, there exists a 2-morphism

$$\alpha\colon \phi \Longrightarrow \psi, \qquad \begin{tikzcd} B \arrow[r, bend left=60, "\phi"{name=U}] \arrow[r, bend right=60, "\psi"'{name=D}] & X \arrow[Rightarrow, from=U, to=D, "\alpha"] \end{tikzcd}$$

of $\mathcal{C}$ such that we have an equality

$$\begin{tikzcd} A \arrow[r, "f"] & B \arrow[r, bend left=60, "\phi"{name=U}] \arrow[r, bend right=60, "\psi"'{name=D}] & X \arrow[Rightarrow, from=U, to=D, "\alpha"] \end{tikzcd} = \begin{tikzcd} A \arrow[r, bend left=60, "\phi\circ f"{name=U}] \arrow[r, bend right=60, "\psi\circ f"'{name=D}] & X \arrow[Rightarrow, from=U, to=D, "\beta"] \end{tikzcd}$$

of pasting diagrams in $\mathcal{C}$, i.e. such that we have

$$\beta = \alpha \star \mathrm{id}_f\,.$$

Example 13.2.2.1.3. Here are some examples of corepresentably full morphisms.

1. *Corepresentably Full Morphisms in* Cats_2. The corepresentably full morphisms in Cats_2 are characterised in Categories, Item 7 of Definition 11.6.2.1.2.

2. *Corepresentably Full Morphisms in* **Rel**. The corepresentably full morphisms in **Rel** are characterised in Relations, Item 2 of Definition 8.4.10.1.1.

13.2.3 Corepresentably Fully Faithful Morphisms

Let $\mathcal{C}$ be a bicategory.

Definition 13.2.3.1.1. A 1-morphism $f\colon A \to B$ of $\mathcal{C}$ is **corepresentably fully faithful**[4] if the following equivalent conditions are satisfied:

[4] *Further Terminology:* Corepresentably fully faithful morphisms have also been called

1. The 1-morphism f is corepresentably full (Definition 13.2.2.1.1) and corepresentably faithful (Definition 13.2.1.1.1).

2. For each $X \in \mathrm{Obj}(\mathcal{C})$, the functor

$$f^* \colon \mathsf{Hom}_{\mathcal{C}}(B, X) \to \mathsf{Hom}_{\mathcal{C}}(A, X)$$

given by precomposition by f is fully faithful.

Remark 13.2.3.1.2. In detail, f is corepresentably fully faithful if the conditions in Definition 13.2.1.1.2 and Definition 13.2.2.1.2 hold:

1. For all diagrams in $\mathcal{C}$ of the form

$$\begin{tikzcd} A \arrow[r, "f"] & B \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\alpha\Downarrow\Downarrow\beta"] & X, \end{tikzcd}$$

if we have

$$\alpha \star \mathrm{id}_f = \beta \star \mathrm{id}_f,$$

then $\alpha = \beta$.

2. For each $X \in \mathrm{Obj}(\mathcal{C})$ and each 2-morphism

$$\beta \colon \phi \circ f \Longrightarrow \psi \circ f, \qquad \begin{tikzcd} A \arrow[r, bend left, "\phi\circ f"] \arrow[r, bend right, "\psi\circ f"'] \arrow[r, phantom, "\beta\Downarrow"] & X \end{tikzcd}$$

of $\mathcal{C}$, there exists a 2-morphism

$$\alpha \colon \phi \Longrightarrow \psi, \qquad \begin{tikzcd} B \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\alpha\Downarrow"] & X \end{tikzcd}$$

of $\mathcal{C}$ such that we have an equality

$$\begin{tikzcd} A \arrow[r, "f"] & B \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\alpha\Downarrow"] & X \end{tikzcd} \quad = \quad \begin{tikzcd} A \arrow[r, bend left, "\phi\circ f"] \arrow[r, bend right, "\psi\circ f"'] \arrow[r, phantom, "\beta\Downarrow"] & X \end{tikzcd}$$

of pasting diagrams in $\mathcal{C}$, i.e. such that we have

$$\beta = \alpha \star \mathrm{id}_f\,.$$

Example 13.2.3.1.3. Here are some examples of corepresentably fully faithful morphisms.

1. *Corepresentably Fully Faithful Morphisms in* Cats_2. The fully faithful epimorphisms in Cats_2 are characterised in Categories, Item 10 of Definition 11.6.3.1.2.

2. *Corepresentably Fully Faithful Morphisms in* $\mathbf{Rel}$. The corepresentably fully faithful morphisms of $\mathbf{Rel}$ coincide (Relations, Item 3 of Definition 8.4.10.1.1) with the corepresentably full morphisms in $\mathbf{Rel}$, which are characterised in Relations, Item 2 of Definition 8.4.10.1.1.

13.2.4 Morphisms Corepresentably Faithful on Cores

Let $\mathcal{C}$ be a bicategory.

Definition 13.2.4.1.1. A 1-morphism $f\colon A \to B$ of $\mathcal{C}$ is **corepresentably faithful on cores** if, for each $X \in \mathrm{Obj}(\mathcal{C})$, the functor

$$f^*\colon \mathsf{Core}(\mathsf{Hom}_{\mathcal{C}}(B, X)) \to \mathsf{Core}(\mathsf{Hom}_{\mathcal{C}}(A, X))$$

given by precomposition by f is faithful.

Remark 13.2.4.1.2. In detail, f is corepresentably faithful on cores if, for all diagrams in $\mathcal{C}$ of the form

$$\begin{tikzcd} A \arrow[r, "f"] & B \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\alpha\Downarrow\Downarrow\beta"] & X, \end{tikzcd}$$

if α and β are 2-isomorphisms and we have

$$\alpha \star \mathrm{id}_f = \beta \star \mathrm{id}_f,$$

then $\alpha = \beta$.

13.2.5 Morphisms Corepresentably Full on Cores

Let $\mathcal{C}$ be a bicategory.

lax epimorphisms in the literature (e.g. in [Adá+01]), though we will always use the name "corepresentably fully faithful morphism" instead in this work.

Definition 13.2.5.1.1. A 1-morphism $f\colon A \to B$ of $\mathcal{C}$ is **corepresentably full on cores** if, for each $X \in \mathrm{Obj}(\mathcal{C})$, the functor

$$f^*\colon \mathsf{Core}(\mathsf{Hom}_{\mathcal{C}}(B, X)) \to \mathsf{Core}(\mathsf{Hom}_{\mathcal{C}}(A, X))$$

given by precomposition by f is full.

Remark 13.2.5.1.2. In detail, f is corepresentably full on cores if, for each $X \in \mathrm{Obj}(\mathcal{C})$ and each 2-isomorphism

$$\beta\colon \phi \circ f \overset{\sim}{\Longrightarrow} \psi \circ f, \qquad \begin{tikzcd} A \arrow[r, bend left, "\phi\circ f"] \arrow[r, bend right, "\psi\circ f"'] \arrow[r, phantom, "\beta\Downarrow"] & X \end{tikzcd}$$

of $\mathcal{C}$, there exists a 2-isomorphism

$$\alpha\colon \phi \overset{\sim}{\Longrightarrow} \psi, \qquad \begin{tikzcd} B \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\alpha\Downarrow"] & X \end{tikzcd}$$

of $\mathcal{C}$ such that we have an equality

$$\begin{tikzcd} A \arrow[r, "f"] & B \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\alpha\Downarrow"] & X \end{tikzcd} = \begin{tikzcd} A \arrow[r, bend left, "\phi\circ f"] \arrow[r, bend right, "\psi\circ f"'] \arrow[r, phantom, "\beta\Downarrow"] & X \end{tikzcd}$$

of pasting diagrams in $\mathcal{C}$, i.e. such that we have

$$\beta = \alpha \star \mathrm{id}_f\,.$$

13.2.6 Morphisms Corepresentably Fully Faithful on Cores

Let $\mathcal{C}$ be a bicategory.

Definition 13.2.6.1.1. A 1-morphism $f\colon A \to B$ of $\mathcal{C}$ is **corepresentably fully faithful on cores** if the following equivalent conditions are satisfied:

1. The 1-morphism f is corepresentably full on cores (Definition 13.2.5.1.1) and corepresentably faithful on cores (Definition 13.2.1.1.1).

2. For each $X \in \mathrm{Obj}(\mathcal{C})$, the functor

$$f^* \colon \mathsf{Core}(\mathsf{Hom}_{\mathcal{C}}(B, X)) \to \mathsf{Core}(\mathsf{Hom}_{\mathcal{C}}(A, X))$$

given by precomposition by f is fully faithful.

Remark 13.2.6.1.2. In detail, f is corepresentably fully faithful on cores if the conditions in Definition 13.2.4.1.2 and Definition 13.2.5.1.2 hold:

1. For all diagrams in $\mathcal{C}$ of the form

$$\begin{tikzcd} A \arrow[r, "f"] & B \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\alpha\Downarrow\Downarrow\beta"] & X, \end{tikzcd}$$

if α and β are 2-isomorphisms and we have

$$\alpha \star \mathrm{id}_f = \beta \star \mathrm{id}_f,$$

then $\alpha = \beta$.

2. For each $X \in \mathrm{Obj}(\mathcal{C})$ and each 2-isomorphism

$$\beta \colon \phi \circ f \overset{\sim}{\Longrightarrow} \psi \circ f, \qquad \begin{tikzcd} A \arrow[r, bend left, "\phi\circ f"] \arrow[r, bend right, "\psi\circ f"'] \arrow[r, phantom, "\beta\Downarrow"] & X \end{tikzcd}$$

of $\mathcal{C}$, there exists a 2-isomorphism

$$\alpha \colon \phi \overset{\sim}{\Longrightarrow} \psi, \qquad \begin{tikzcd} B \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\alpha\Downarrow"] & X \end{tikzcd}$$

of $\mathcal{C}$ such that we have an equality

$$\begin{tikzcd} A \arrow[r, "f"] & B \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\alpha\Downarrow"] & X \end{tikzcd} = \begin{tikzcd} A \arrow[r, bend left, "\phi\circ f"] \arrow[r, bend right, "\psi\circ f"'] \arrow[r, phantom, "\beta\Downarrow"] & X \end{tikzcd}$$

of pasting diagrams in $\mathcal{C}$, i.e. such that we have

$$\beta = \alpha \star \mathrm{id}_f\,.$$

13.2.7 Corepresentably Essentially Injective Morphisms

Let $\mathcal{C}$ be a bicategory.

Definition 13.2.7.1.1. A 1-morphism $f\colon A \to B$ of $\mathcal{C}$ is **corepresentably essentially injective** if, for each $X \in \mathrm{Obj}(\mathcal{C})$, the functor

$$f^*\colon \mathsf{Hom}_{\mathcal{C}}(B, X) \to \mathsf{Hom}_{\mathcal{C}}(A, X)$$

given by precomposition by f is essentially injective.

Remark 13.2.7.1.2. In detail, f is corepresentably essentially injective if, for each pair of morphisms $\phi, \psi\colon B \rightrightarrows X$ of $\mathcal{C}$, the following condition is satisfied:

($\star$) If $\phi \circ f \cong \psi \circ f$, then $\phi \cong \psi$.

13.2.8 Corepresentably Conservative Morphisms

Let $\mathcal{C}$ be a bicategory.

Definition 13.2.8.1.1. A 1-morphism $f\colon A \to B$ of $\mathcal{C}$ is **corepresentably conservative** if, for each $X \in \mathrm{Obj}(\mathcal{C})$, the functor

$$f^*\colon \mathsf{Hom}_{\mathcal{C}}(B, X) \to \mathsf{Hom}_{\mathcal{C}}(A, X)$$

given by precomposition by f is conservative.

Remark 13.2.8.1.2. In detail, f is corepresentably conservative if, for each pair of morphisms $\phi, \psi\colon B \rightrightarrows X$ and each 2-morphism

$$\alpha\colon \phi \overset{\sim}{\Longrightarrow} \psi, \qquad \begin{tikzcd} B \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\alpha\Downarrow"] & X \end{tikzcd}$$

of $\mathcal{C}$, if the 2-morphism

$$\alpha \star \mathrm{id}_f\colon \phi \circ f \Longrightarrow \psi \circ f, \qquad \begin{tikzcd} A \arrow[r, bend left, "\phi\circ f"] \arrow[r, bend right, "\psi\circ f"'] \arrow[r, phantom, "\alpha\star\mathrm{id}_f\Downarrow"] & X \end{tikzcd}$$

is a 2-isomorphism, then so is α.

13.2.9 Strict Epimorphisms

Let $\mathcal{C}$ be a bicategory.

Definition 13.2.9.1.1. A 1-morphism $f\colon A \to B$ is a **strict epimorphism in $\mathcal{C}$** if, for each $X \in \mathrm{Obj}(\mathcal{C})$, the functor

$$f^*\colon \mathsf{Hom}_{\mathcal{C}}(B, X) \to \mathsf{Hom}_{\mathcal{C}}(A, X)$$

given by precomposition by f is injective on objects, i.e. its action on objects

$$f_*\colon \mathrm{Obj}(\mathsf{Hom}_{\mathcal{C}}(B, X)) \to \mathrm{Obj}(\mathsf{Hom}_{\mathcal{C}}(A, X))$$

is injective.

Remark 13.2.9.1.2. In detail, f is a strict epimorphism if, for each diagram in $\mathcal{C}$ of the form

$$A \xrightarrow{f} B \underset{\psi}{\overset{\phi}{\rightrightarrows}} X,$$

if $\phi \circ f = \psi \circ f$, then $\phi = \psi$.

Example 13.2.9.1.3. Here are some examples of strict epimorphisms.

1. *Strict Epimorphisms in* Cats_2. The strict epimorphisms in Cats_2 are characterised in Categories, Item 1 of Definition 11.7.3.1.2.

2. *Strict Epimorphisms in* $\mathbf{Rel}$. The strict epimorphisms in $\mathbf{Rel}$ are characterised in Relations, Definition 8.4.9.1.1.

13.2.10 Pseudoepic Morphisms

Let $\mathcal{C}$ be a bicategory.

Definition 13.2.10.1.1. A 1-morphism $f\colon A \to B$ of $\mathcal{C}$ is **pseudoepic** if, for each $X \in \mathrm{Obj}(\mathcal{C})$, the functor

$$f^*\colon \mathsf{Hom}_{\mathcal{C}}(B, X) \to \mathsf{Hom}_{\mathcal{C}}(A, X)$$

given by precomposition by f is pseudomonic.

Remark 13.2.10.1.2. In detail, a 1-morphism $f\colon A \to B$ of $\mathcal{C}$ is pseudoepic if it satisfies the following conditions:

1. For all diagrams in $\mathcal{C}$ of the form

$$\begin{tikzcd} A \arrow[r, "f"] & B \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\alpha\Downarrow\Downarrow\beta"] & X, \end{tikzcd}$$

   if we have

$$\alpha \star \mathrm{id}_f = \beta \star \mathrm{id}_f,$$

   then $\alpha = \beta$.

2. For each $X \in \mathrm{Obj}(\mathcal{C})$ and each 2-isomorphism

$$\beta\colon \phi \circ f \overset{\sim}{\Longrightarrow} \psi \circ f, \qquad \begin{tikzcd} A \arrow[r, bend left, "\phi\circ f"] \arrow[r, bend right, "\psi\circ f"'] \arrow[r, phantom, "\beta\Downarrow"] & X \end{tikzcd}$$

   of $\mathcal{C}$, there exists a 2-isomorphism

$$\alpha\colon \phi \overset{\sim}{\Longrightarrow} \psi, \qquad \begin{tikzcd} B \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\alpha\Downarrow"] & X \end{tikzcd}$$

   of $\mathcal{C}$ such that we have an equality

$$\begin{tikzcd} A \arrow[r, "f"] & B \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\alpha\Downarrow"] & X \end{tikzcd} \quad = \quad \begin{tikzcd} A \arrow[r, bend left, "\phi\circ f"] \arrow[r, bend right, "\psi\circ f"'] \arrow[r, phantom, "\beta\Downarrow"] & X \end{tikzcd}$$

   of pasting diagrams in $\mathcal{C}$, i.e. such that we have

$$\beta = \alpha \star \mathrm{id}_f\,.$$

Proposition 13.2.10.1.3. Let $f\colon A \to B$ be a 1-morphism of $\mathcal{C}$.

1. *Characterisations.* The following conditions are equivalent:

   (a) The morphism f is pseudoepic.

   (b) The morphism f is corepresentably full on cores and corepresentably faithful.

(c) We have an isococomma square of the form

$$B \overset{\text{eq.}}{\cong} B \overset{\leftrightarrow}{\coprod}_A B, \qquad \begin{tikzcd} B & B \arrow[l, "\mathrm{id}_B"'] \\ B \arrow[u, "\mathrm{id}_B"] & A \arrow[l, "F"] \arrow[u, "F"'] \arrow[ul, leftrightarrow, dashed] \end{tikzcd}$$

in $\mathcal{C}$ up to equivalence.

Proof. *Item 1*, *Characterisations*: Omitted. □

Appendices

A Other Chapters

Preliminaries

1. Introduction
2. A Guide to the Literature

Sets

3. Sets
4. Constructions With Sets
5. Monoidal Structures on the Category of Sets
6. Pointed Sets
7. Tensor Products of Pointed Sets

Relations

8. Relations
9. Constructions With Relations
10. Conditions on Relations

Categories

11. Categories

Monoidal Categories

12. Constructions With Monoidal Categories

Bicategories

13. Types of Morphisms in Bicategories

Extra Part

14. Notes

References

[Adá+01] Jiří Adámek, Robert El Bashir, Manuela Sobral, and Jiří Velebil. "On Functors Which Are Lax Epimorphisms". In: *Theory Appl. Categ.* 8 (2001), pp. 509–521. ISSN: 1201-561X (cit. on p. 15).